



Lakshya JEE 2.0 (2024)

Determinants

DPP-02

1. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, then value of $\begin{vmatrix} 2a_1+3b_1+4c_1 & b_1 & c_1 \\ 2a_2+3b_2+4c_2 & b_2 & c_2 \\ 2a_3+3b_3+4c_3 & b_3 & c_3 \end{vmatrix}$ is

equal to

- (1) Δ^2 (2) 4Δ
 (3) Δ (4) 2Δ

2. The parameter, on which the value of the determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

does not depend upon, is:

- (1) a (2) p
 (3) d (4) x

3. The value of the determinant $\begin{vmatrix} {}^5C_0 & {}^5C_3 & 14 \\ {}^5C_1 & {}^5C_4 & 1 \\ {}^5C_2 & {}^5C_5 & 1 \end{vmatrix}$ is:

- (1) 0 (2) $-(6!)$
 (3) 80 (4) -576

4. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = ka^2b^2c^2$, then k is equal to:

- (1) 2 (2) 4
 (3) -4 (4) 8

5. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$, then the

 ordered pair (A, B) is equal to

- (1) $(4, 5)$ (2) $(-4, -5)$
 (3) $(-4, 3)$ (4) $(-4, 5)$

6. $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix} =$

(1) $xyz \left(1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$

(2) xyz

(3) $1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

(4) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

7. The value of $\Delta = \begin{vmatrix} {}^{10}C_4 & {}^{10}C_5 & {}^{11}C_m \\ {}^{11}C_6 & {}^{11}C_7 & {}^{12}C_{m+2} \\ {}^{12}C_8 & {}^{12}C_9 & {}^{13}C_{m+4} \end{vmatrix}$ is equal to

 zero, where m is

- (1) 6 (2) 4
 (3) 5 (4) None of these

8. The value of

$$\begin{vmatrix} 1 & 1 & 1 \\ (2^x + 2^{-x})^2 & (3^x + 3^{-x})^2 & (5^x + 5^{-x})^2 \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix}$$

- (1) 0 (2) 30^x
 (3) 30^{-x} (4) None of these

9. If x, y, z are integers in A.P. lying between 1 and 9 and $x51, y41$ and $z31$ are three digit numbers then

the value of $\begin{vmatrix} 5 & 4 & 3 \\ x51 & y41 & z31 \\ x & y & z \end{vmatrix}$ is

- (1) $x + y + z$ (2) $x - y + z$
 (3) 0 (4) None of these



10. If $a \neq b \neq c$, the value of x which satisfies the

$$\text{equation } \begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0 \text{ is}$$

- (1) $x = 0$ (2) $x = a$
(3) $x = b$ (4) $x = c$

11. The value of $\begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 4 & 4 & 3 & 0 & 0 \\ 5 & 5 & 5 & 4 & 0 \\ 6 & 6 & 6 & 6 & 5 \end{vmatrix}$ is

- (1) $6!$ (2) $5!$
(3) $1 \cdot 2^2 \cdot 3 \cdot 4^3 \cdot 5^4 \cdot 6^4$ (4) None of these

12. The cofactor of the element 4 in the determinant

$$\begin{vmatrix} 1 & 3 & 5 & 1 \\ 2 & 3 & 4 & 2 \\ 8 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \end{vmatrix} \text{ is}$$

- (1) 4 (2) 10
(3) -10 (4) -4

13. $\begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix}$ is equal to

- (1) abc (2) $2abc$
(3) $4abc$ (4) 0

14. $\begin{vmatrix} 2^3 & 3^3 & 3 \cdot 2^2 + 3 \cdot 2 + 1 \\ 3^3 & 4^3 & 3 \cdot 3^2 + 3 \cdot 3 + 1 \\ 4^3 & 5^3 & 3 \cdot 4^2 + 3 \cdot 4 + 1 \end{vmatrix}$ is equal to

- (1) 0 (2) 1
(3) 92 (4) None of these

15. If $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = \lambda \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ then λ is equal

to

- (1) 1 (2) 2
(3) 3 (4) 4

16. If $\Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$ then Δ_1 is

equal to

- (1) $2\Delta_2$ (2) Δ_2
(3) $-\Delta_2$ (4) None of these

17. If $\Delta_1 = \begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \end{vmatrix}$ and

$$\Delta_2 = \begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} \text{ then}$$

- (1) $\Delta_1 = 2\Delta_2$ (2) $\Delta_2 = 2\Delta_1$
(3) $\Delta_1 = \Delta_2$ (4) None of these

18. If ω, ω^2 are imaginary cube roots of unity and

$$\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{vmatrix} \text{ and } \Delta_2 = \begin{vmatrix} 1 & 1 & \omega \\ 1 & 1 & \omega^2 \\ \omega^2 & \omega & 1 \end{vmatrix}, \text{ then } \frac{\Delta_1}{\Delta_2}$$

is equal to

- (1) $\sqrt{3}$ (2) $\sqrt{3}i$
(3) 1 (4) -1



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (2)
3. (4)
4. (2)
5. (4)
6. (1)
7. (3)
8. (1)
9. (3)

10. (1)
11. (2)
12. (2)
13. (3)
14. (1)
15. (2)
16. (2)
17. (2)
18. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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